The Decision Value of Hidden Supply-Network Structure: Zero-Value Conditions and Identification-Premium Bounds

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Abstract

Firms invest in sub-tier supply-network visibility through mapping, supplier questionnaires and audits, and a substantial literature studies how hidden supplier structure can be inferred. We study a prior question: what is that structure worth to a specific mitigation decision? We formulate discovery-coupled mitigation as a probe-then-secure problem, with no probe cost priced here, on a partially observed supply directed acyclic graph and analyse the expected value of perfect structural information (EVPI). We give sufficient structural conditions, namely invariance of the loss distribution under identification-free plans together with uniform plan-level dominance of identification-free actions, under which EVPI is exactly zero; since perfect information dominates any acquisition policy, every feasible information-acquisition policy then has zero decision value within the modelled problem. The conditions are stated for any monotone law-invariant objective, so they cover expected loss and conditional value at risk alike. Outside that regime we bound EVPI above by a sum of per-category identification premiums computable from prior and public data, prove the bound for node-disjoint categories, and give the premium exactly in a dilution special case, together with a sub-case of it in which the bound is attained. A sufficient condition for the harder of the two hypotheses is read off the graph and the cost vector, and we report how often it holds on real topologies. An empirical study on 32 benchmark multi-echelon topologies derived from the Willems dataset, with model-generated latent overlays, measures the advantage that perfect structural information confers on an implemented planner. That advantage is small under the loss normalisation used here, with a median of exactly zero in the public-attribute regime, and a census of the same chains finds that **88.9%** of their categories have a single possible supplier, which bounds how much there was to find. We give the exact relationship between the measured quantity and EVPI and do not claim it bounds EVPI from either side.

Keywords: value of information, EVPI, supply network visibility, disruption risk, conditional value at risk, endogenous uncertainty

1 Introduction

Sub-tier supply-network visibility is bought through mapping services, supplier questionnaires, audit programmes and related discovery activities. The case for such spending is usually argued from the operational consequences of incomplete visibility. It is much less common to price the information itself: to ask by how much a firm's mitigation decisions would improve if the hidden structure were supplied at no cost.

Decision analysis answers that question with the expected value of perfect information, which bounds the improvement obtainable from any amount of investigation. Applying it to supply-network structure is easy to state and harder to use, because the decision the information feeds — which suppliers to qualify a backup for, under a cost budget, on a network whose upstream shape is the unknown — is combinatorial.

We formulate this problem as a decision model of discovery-coupled mitigation and characterise its information value. One point of scope should be clear from the outset: we price the information, not the coupling. No probe cost and no shared probe-and-mitigation budget enters any quantity reported here, since the uninformed planner acts on the prior and the informed planner is handed θ outright. The first result is negative and, in our view, the more useful one: there are structural conditions under which the value of perfect structural information is exactly zero. Both can be examined before any structural discovery is commissioned: the first, a symmetry of the public network that exchanges the candidates of each category together with their consumers, upstream subtrees and correlation clusters, directly from public data, and the second through a sufficient condition on the graph and the cost vector. The result is independent of the optimisation algorithm used to implement the decision: under those conditions no information-acquisition policy can improve the objective, because perfect information dominates every such policy and is itself worth nothing. The second result is a bound for the cases where those conditions fail, expressed as a sum of per-category identification premiums, each computable from information the firm already holds.

Central claim.

We characterise sufficient structural conditions under which the decision value of hidden supplier information is exactly zero, and derive a computable upper bound on that value outside the zero-value regime. Everything else in the paper supports that claim: the screening score is secondary, the enumeration experiments are verification, the CVaR statement is an extension, and the shared-subtree case is an unresolved extension we label as such.

Contributions.

1. *Zero-value characterisation.* Theorem 4 gives sufficient conditions for EVPI = 0, and Lemma 2 transfers the conclusion to every information-acquisition policy. Proposition 8 gives a sufficient condition for the harder of the two hypotheses that is read off the public graph and cost vector, and Section 7.3 reports how often it holds on the benchmark collection. Corollary 5 extends the statement to monotone law-invariant objectives, covering conditional value at risk.
2. *Identification-premium upper bound.* Theorem 6 bounds EVPI by a finite, directly computable sum of per-category premiums for node-disjoint categories, and Corollary 9 bounds the partial value of each category separately. Computable EVPI bounds are not new (Section 2, stream A′). What we believe to be new, and have not found stated elsewhere, is a bound of this form: it requires no convexity, decomposes over a discrete latent assignment, and is localised to individual channels so that they can be screened separately. Further, Corollary 7 gives the exact premium in a dilution special case, and we identify a sub-case of it in which the bound is attained with equality; and Proposition 10 gives the premium in closed form under a specified dispersion family, identifying a threshold below which dispersion does not enter the bound at all.
3. *Computational verification.* Exhaustive enumeration of exact miniature instances tests the implementation of the definitions and searches for violations that would reveal errors in the stated results (Section 6).
4. *Empirical characterisation.* Measurements on benchmark multi-echelon topologies and a synthetic family, across a public-attribute and a latent-attribute regime (Section 7).

Section 8 states what the paper does not establish.

2 Related work and positioning

(A) Value of information.

The expected value of perfect information originates with Raiffa and Schlaifer [22] and Howard [15]. Its partial form, EVPPI, is standard in health economics, where the computational cost of nested estimation is a recognised difficulty [14, 26]. Our premium is an EVPPI-shaped quantity for structural parameters of a network, evaluated by finite enumeration over budget breakpoints rather than by nested simulation. We claim no priority over that literature, whose machinery we reuse.

(A′) Bounding EVPI when it cannot be computed.

We must be explicit that a computable upper bound on EVPI is not a new idea. Because exact evaluation is often prohibitive, bounds have been derived since Madansky [21], and Avriel and Williams [5] obtained EVPI bounds evaluable as part of a post-optimal analysis of a linear or convex program. Ziemba and Butterworth [31] and Hausch and Ziemba [13] devote a two-part study to bounding the value of information directly; Huang, Vertinsky and Ziemba [16] sharpen the perfect-information bounds for simple-recourse problems; and Birge [6] develops the companion bounds for the

value of the stochastic solution. That family of bounds rests on convexity or concavity: Huang, Vertinsky and Ziemba state theirs for simple-recourse problems with nonlinear return functions and concave utility functions, using extensions of the Jensen and Edmundson–Madansky inequalities. Theorem 6 uses no convexity and no continuity: it is a combinatorial decomposition over a discrete latent assignment, evaluated on a finite set of budget breakpoints, and it localises the bound to individual categories so that channels can be screened separately. Our contribution is that specific form and its use for screening, not the general idea of bounding EVPI, and where the two apply to the same problem we make no claim that ours is tighter.

(B) Information acquisition and active learning.

Adaptive information gathering with performance guarantees is well developed [12], and decision-coupled acquisition, which drives uncertainty into a single decision region instead of reducing it uniformly, is the closest precedent for the objective used here [17]. The ordering of information structures by their decision value is Blackwell's [7], and Lemma 2 is a standard consequence of it rather than a contribution of this paper. That literature asks how to acquire information efficiently. We ask when acquisition cannot help at all, which is logically prior and is not settled by any performance guarantee.

(C) Decision-dependent information discovery.

Stochastic programming with endogenous uncertainty studies problems in which decisions determine what is revealed [11, 18], and the robust counterpart optimises under decision-dependent revelation [28]. Artstein and Wets [1] and Artstein [2, 3] treat the information structure itself as a decision variable through the notion of sensors; Artstein [2] recasts a two-stage problem with a probing decision as a three-stage stochastic program and establishes a strong law of large numbers for it, and Artstein [3] addresses the trade-off between the gain from new information and the cost of obtaining it. Closest to our setting is the probing-enhanced stochastic program of Ma et al. [20], in which costly discrete probes are chosen before a two-stage decision and the perfect-information bound is the benchmark to be improved upon; their contribution is algorithmic, a branch-and-bound scheme over probing sets. The direction of the question differs: that stream computes policies *under* revelation, while we characterise conditions in which revelation changes nothing, so that the problem those methods solve has a degenerate answer.

(D) Valuation of supply-chain visibility.

It would be wrong to claim that visibility has not been valued. Caridi et al. [9] develop a structured value-assessment model for visibility and apply it to a real chain, and recent student and practitioner work continues to estimate returns to deeper tiers [10]. That work maps visibility improvements to operational and strategic benefits, typically through causal models of performance indicators. Our question is narrower and its answer sharper: for a specified mitigation decision and a specified loss, when is the decision value of latent structure exactly zero, and when it is not, what upper bound can be computed in advance. Neither statement subsumes the other.

(E) Multi-tier structure and hidden links.

Inference of hidden supplier relationships is an active area [8, 19], and topological measures of network vulnerability are well established [24, 29]. These methods find or score structure; they do not price it against a decision.

(F) Disruption mitigation.

Mitigation and contingency strategy follows Tomlin [27], with the wider OR/MS literature surveyed by Snyder et al. [25]. Our mitigation model — budgeted backup qualification under a conjunctive cascade — is deliberately simple, since the object of study is the information rather than the mitigation policy.

(G) Risk measures.

The tail machinery is that of Rockafellar and Uryasev [23] and Artzner et al. [4].

Table 1 summarises where this paper sits against these streams. The column that separates it is the third.

Table 1 Positioning against adjacent streams. The column that separates this paper is the sufficient zero-value condition; computable EVPI bounds already exist (stream A′), and our claim there concerns the form of the bound rather than its existence. The zero-value column records what we found in the works we examined, not an exhaustive claim about each stream.

stream	values latent structure	gives a zero-value condition	gives an EVPI bound	bound is convexity-free	applied to supply networks
Value of information (A)	yes	no	no	n/a	no
EVPI bounding (A′)	no	no	yes	not in those examined	no
Active acquisition (B)	yes	no	no	n/a	no
Decision-dependent discovery (C)	yes	no	no	n/a	no
Visibility valuation (D)	yes	no	no	n/a	yes
Hidden-link inference (E)	no	no	no	n/a	yes
This paper	yes	yes	yes	yes	yes

3 Model

Network.

A finite directed acyclic graph $G = (V, E)$, where $u \to v$ means u supplies v. A focal node 0 consumes K_0 input categories. Every node v consumes categories $k \in K_v$, each served by a set of suppliers. Service is conjunctive across categories and substitutable within a category, in the sense that any one working supplier suffices. That clause fixes the semantics of the cascade rather than the sourcing structure. The candidate set is epistemic rather than a redundancy: exactly one candidate actually supplies

the category, and the set records who it might be. The model therefore contains no concurrently multi-sourced category, and dual sourcing as a mitigation lies outside it. Each node has a publicly known securing cost c_v and belongs to a publicly known correlation cluster. A node's *tier* is its distance from the focal node along the supply direction: the focal firm's direct suppliers are tier one, their suppliers tier two, and so on.

Uncertainty.

The structure parameter θ specifies, for each latent category (v, k), which candidate in a finite set $C_{v,k}$ is the true supplier, exactly one candidate serving the category, an assumption Corollary 7 and Proposition 10 use and, in the dispersed regime of Section 7, which candidate carries which risk attribute. A prior π over θ is given. Probing a node reveals its candidate in-edges through a noisy asymmetric channel. That channel belongs to the wider model, of which this paper studies only the value of the information a probe could deliver; it enters neither EVPI nor the quantity measured in Section 7, since the uninformed planner sees only the prior and the informed planner sees θ outright.

Disruption and loss.

A scenario ω draws node-level failures, correlated within clusters by a one-factor Gaussian copula. Operability propagates through the conjunctive cascade. A plan $S \subseteq V \setminus \{0\}$ with $c(S) \le B$ qualifies backup sources: $u \in S$ guarantees u's output to its consumers. The focal loss is the fraction of focal categories starved,

$$L(S;\theta,\omega) = \frac{1}{K_0}\sum_{k=1}^{K_0} Z_k, \qquad Z_k = \mathbf{1}\{\text{focal category } k \text{ starved}\}. \tag{1}$$

Objectives and the exact target.

Write $j(S;\theta) = \mathbb{E}_\omega L(S;\theta,\omega)$ and $J(S;\theta) = \mathrm{CVaR}_{\alpha,\omega}(L(S;\theta,\omega))$. Let S_π minimise $\mathbb{E}_{\theta\sim\pi} j(S;\theta)$ and S_θ minimise $j(S;\theta)$. Then

$$\mathrm{EVPI}(\pi) = \mathbb{E}_\pi[j(S_\pi;\theta)] - \mathbb{E}_\pi[j(S_\theta;\theta)] \;\ge\; 0. \tag{2}$$

Units.

L is a fraction of focal categories, so every quantity here is dimensionless. Because L is normalised by K_0, and K_0 runs from 2 to 672 across the retained chains, the same absolute value means very different things on different chains: rescuing one focal category moves the objective by $1/K_0$, which is 0.019 at the median $K_0 = 52$ and 0.0015 at $K_0 = 672$. An advantage of 0.0011 on a chain at the median K_0 would thus be about a twentieth of one focal category. That is an illustration at one value of K_0, not the measurement. The measurement is the scaled quantity $H \cdot K_0$, which says directly how many focal categories perfect structural knowledge rescues: across the benchmark instances it averages 0.020 at $\lambda = 0$, with a median of exactly zero and a maximum of 1.012, and 0.146 at $\lambda = 0.8$, with a median of 0.018 and a maximum of 2.379. Perfect

knowledge therefore rescues on average a fiftieth of one focal category in the public-attribute regime and about a seventh of one in the latent-attribute regime, and at most 1.01 and 2.38 respectively. The illustration and the measurement differ by about a factor of three at $\lambda = 0$ because H is smaller on the chains with large K_0; Section 7.6 reports how the scaled quantity varies with K_0. A value of 0.005 means that perfect structural knowledge would reduce the objective by half a percentage point of the focal basket: expected category starvation where the objective is j, and the conditional tail average of the starved fraction where it is J, as it is in the measurements of Section 7. Whether that is a large amount of money depends on what one starved category costs the firm, which this model does not represent; Section 7.7 gives the conversion a reader needs to price it for a specific firm.

3.1 Three quantities that must not be conflated

The empirical sections do not measure (2). A firm operates an implemented planner rather than an exact optimiser, and so do we. Let $\hat{S}_\pi$ and $\hat{S}_\theta$ be the plans returned by the implemented planner given the prior and given θ. Define the *heuristic oracle advantage*

$$H \;=\; \mathbb{E}_\pi\Big[j(\hat{S}_\pi;\theta)\Big] - \mathbb{E}_\pi\Big[j(\hat{S}_\theta;\theta)\Big]\,. \tag{3}$$

Proposition 1 (Exact relation between H and EVPI) *Let* $\varepsilon_\pi = \mathbb{E}_\pi[j(\hat{S}_\pi;\theta)] - \mathbb{E}_\pi[j(S_\pi;\theta)] \geq 0$ *and* $\varepsilon_\theta = \mathbb{E}_\pi[j(\hat{S}_\theta;\theta) - j(S_\theta;\theta)] \geq 0$ *be the planner's optimality gaps in the two regimes. Then*

$$H = \mathrm{EVPI}(\pi) + \varepsilon_\pi - \varepsilon_\theta. \tag{4}$$

Hence $H \leq \mathrm{EVPI}$ *if and only if* $\varepsilon_\pi \leq \varepsilon_\theta$*. In general the sign of* $\varepsilon_\pi - \varepsilon_\theta$ *is not determined, so* H *is neither an upper nor a lower bound on EVPI; if the planner is* θ*-optimal when given* θ*, so that* $\varepsilon_\theta = 0$*, then* $H \geq \mathrm{EVPI}$*.*

Proof (E1) By definition of ε_π, $\mathbb{E}[j(\hat{S}_\pi;\theta)] = \mathbb{E}[j(S_\pi;\theta)] + \varepsilon_\pi$.
(E2) By definition of ε_θ, $\mathbb{E}[j(\hat{S}_\theta;\theta)] = \mathbb{E}[j(S_\theta;\theta)] + \varepsilon_\theta$.
(E3) Subtracting (E2) from (E1) and using (2) gives (4). Both gaps are nonnegative because S_π and S_θ minimise over the same feasible sets that the planner searches heuristically, and neither gap dominates the other in general, which gives the remaining statements. □

Proposition 1 holds for any fixed objective, provided the same one is used in every term. We fix the convention now, because the paper uses two. The measurements of Section 7 take J, the CVaR objective, throughout: H, $\hat{S}_\pi$ and $\hat{S}_\theta$ there are CVaR quantities, and the EVPI they relate to by (4) is the CVaR value of perfect information, not (2). The enumeration of Section 7.8 takes j, expected loss, so that its EVPI is (2) exactly. The premium bound U is stated for expected loss only and is never applied to a CVaR measurement anywhere in this paper. Given that convention, H is a planner-specific measurement and is reported as such throughout. Section 7.8 measures both gaps directly on instances where exact optimisation is possible, and reports how large the discrepancy actually is and how its sign behaves. Table 2 fixes the vocabulary.

Table 2 The four quantities used in this paper.

quantity	definition	exact?	direction
EVPI	prior-optimal versus state-optimal plan, exact optimisation	exact	the target
$U = K_0^{-1} \sum_k I_k$	sum of identification premiums, Thm. 6	exact upper bound	EVPI $\leq U$ under the stated assumptions
H	same implemented planner with and without θ	approximate	$H = \text{EVPI} + \varepsilon_\pi - \varepsilon_\theta$; sign undetermined
SISS	screening score, App. A	heuristic	no bound claimed outside Thm. 6

3.2 Units of analysis

Several nested objects appear in the empirical section, so we fix them here. A *network* is one benchmark topology. An *overlay* is one seeded assignment of latent structure and risk attributes to a network. An *instance* is a network–overlay pair and is the unit at which H is computed. A *channel* is one latent category at one node and is the unit at which premiums and screening scores are computed. A *focal category* is one of the K_0 inputs the focal node consumes. Summary statistics of H are computed at instance level; reliability is assessed at network level, because overlays of the same network are not independent replicates.

4 When structural information has zero value

Lemma 2 (Acquisition is dominated by perfect information) *For any information-acquisition policy φ with value* $\text{Val}(\varphi)$, $\text{Val}(\varphi) \leq \text{EVPI}(\pi)$.

Proof (L1) By definition, $\text{Val}(\varphi) = \mathbb{E}_\pi[j(S_\pi;\theta)] - \mathbb{E}_o\mathbb{E}_{\theta|o}[j(S_{\pi_o};\theta)]$, where o is the observation induced by φ, π_o the posterior and S_{π_o} the plan chosen after o.
(L2) Pointwise in (o,θ), S_{π_o} is one feasible plan among others, so $j(S_{\pi_o};\theta) \geq \min_S j(S;\theta) = j(S_\theta;\theta)$.
(L3) Taking $\mathbb{E}_{\theta|o}$ of (L2), then $\mathbb{E}_o$, and applying the tower property gives $\mathbb{E}_o\mathbb{E}_{\theta|o}[j(S_{\pi_o};\theta)] \geq \mathbb{E}_\theta[j(S_\theta;\theta)]$.
(L4) Substituting (L3) into (L1) yields $\text{Val}(\varphi) \leq \text{EVPI}(\pi)$. □

Proposition 3 (Bernoulli CVaR) *For $Z \in \{0,1\}$ with $\mathbb{P}(Z=1) = p$,* $\text{CVaR}_\alpha(Z) = \min\{1, p/(1-\alpha)\}$, *and* $\text{CVaR}_\alpha(K_0^{-1}\sum_k Z_k) \leq K_0^{-1}\sum_k \min\{1, p_k/(1-\alpha)\}$.

Proof (P1) If $p \geq 1-\alpha$ the worst $(1-\alpha)$ tail consists of ones only, so the tail average is 1.
(P2) If $p < 1-\alpha$ the tail contains all p mass of ones and $(1-\alpha-p)$ of zeros, so its average is $p/(1-\alpha)$.
(P3) Combining gives $\text{CVaR}_\alpha(Z) = \min\{1, p/(1-\alpha)\}$.

(P4) CVaR is subadditive and positively homogeneous [4], so $\mathrm{CVaR}_\alpha(K_0^{-1}\sum_k Z_k) \le K_0^{-1}\sum_k \mathrm{CVaR}_\alpha(Z_k)$, and (P3) gives the stated form. □

Definition 1 (Identification-free actions) An action is *identification-free* for the latent category (v,k) if the conditional law of the focal loss L, given a plan containing that action, does not depend on $\theta_{v,k}$. A plan is *identification-free* if each of its elements is identification-free for the category it addresses. This is a condition on the plan's own elements only: it says nothing about the latent categories the plan leaves untouched, whose θ-dependence is what (H1) below rules out. Canonical examples of identification-free actions: secure the consumer v; secure every candidate in $C_{v,k}$; and, under the sufficient condition (H1′) below, take no action, since every structure automorphism fixes the empty plan. Under heterogeneous latent candidate risk the last of these is not identification-free, and exchangeability of the candidates' marginal laws alone does not make it so, since the law of L under inaction can depend on which candidate is the supplier. Securing one named candidate $u \in C_{v,k}$ is *identification-dependent*, since its effect depends on whether u is the supplier.

We use two hypotheses.

(H1) *Loss-law invariance.* For every identification-free plan S, the law of $L(S;\theta,\cdot)$ is the same for all θ. The hypothesis is not implied by Definition 1: an identification-free plan constrains only the categories its own elements address, and (H1) is exactly the requirement that the categories it leaves alone contribute a law that does not vary with θ. That is why the sufficient condition (H1′) below is stated over every latent category rather than over the plan's. The experimental construction of Section 7 makes candidates approximately exchangeable in marginal law at $\lambda = 0$, which is not sufficient for (H1), and deliberately breaks even that at $\lambda = 0.8$.

(H2) *Uniform plan-level dominance.* Every identification-dependent element u of any plan admits an identification-free replacement $a(u)$ with $c(a(u)) \le c(u)$ and $j(S\setminus\{u\}\cup\{a(u)\};\theta) \le j(S;\theta)$ for all θ. Securing every candidate in the category is identification-free but cannot serve as $a(u)$ under positive costs: the replacement set contains u, so it is never cheaper. Only an action at the consumer level, or outside the category, can witness the cost clause.

A checkable sufficient condition for (H1).

Exchangeability of the candidates of each category in marginal operability law, in securing cost and under the prior is *not* sufficient. Two candidates with the same failure law can differ in their other consumers, in their upstream subtrees or in their correlation cluster, and each of these differences can make the joint law of the starvation indicators, and with it the law of L, depend on which candidate is the supplier. Example 1 in Appendix C gives an instance in which the marginal condition holds, (H2) holds by Proposition 8, and EVPI > 0. What suffices is a symmetry of the public structure. Write q_v for the marginal failure probability of node v and $\kappa(v)$ for its cluster label. A *structure automorphism* is a bijection $\sigma : V \to V$, together with bijections $\sigma_v : K_v \to K_{\sigma v}$ and a bijection η of cluster labels, such that (A0) $\sigma(0) = 0$; (A1) whenever (v,k) has public supplier s, $(\sigma v, \sigma_v k)$ has public supplier $\sigma(s)$; (A2) (v,k) is latent if and only if $(\sigma v, \sigma_v k)$ is, and then $C_{\sigma v,\sigma_v k} = \sigma(C_{v,k})$; (A3) $q_{\sigma v} = q_v$,

$\kappa(\sigma v) = \eta(\kappa(v))$, and η preserves each cluster's copula parameter. Because σ preserves every supply and candidate relation, it carries each node's upstream subtree and consumers along with the node. It acts on states by $(\sigma\theta)_{\sigma v, \sigma_v k} = \sigma(\theta_{v,k})$ and on plans by $\sigma S = \{\sigma u : u \in S\}$.

(H1′) *Automorphic exchangeability.* For every identification-free plan S and all θ, θ' there is a structure automorphism σ with $\sigma\theta = \theta'$ and $\sigma S = S$.

Lemma 13 in Appendix C proves that (H1′) implies (H1); in fact the joint law of $(Z_1, \ldots, Z_{K_0})$ is then invariant in θ up to a fixed permutation of the focal categories. Neither the prior nor the cost vector enters, since (H1) concerns a fixed plan pointwise in θ. A local form is easier to check. It suffices that for every latent category (v, k) and all $u, u' \in C_{v,k}$ some structure automorphism maps u to u', fixes v and k, fixes pointwise every other candidate set and every other latent category's consumer, and fixes every identification-free plan. The typical witness swaps the upstream cones of two candidates when the cones are disjoint, isomorphic with matching failure probabilities and cluster labels, and joined to the rest of the network only by the edge into (v, k). A candidate that belongs to two candidate sets, or a candidate with a consumer that its rivals lack, defeats the local form.

Theorem 4 (Zero decision value of hidden structure) *Under (H1) and (H2),* $\mathrm{EVPI}(\pi) = 0$*; hence by Lemma 2 every feasible information-acquisition policy has zero decision value.*

Proof (T1) Fix θ. The feasible plan set is finite, so a θ-optimal plan S_θ exists.
(T2) Suppose S_θ contains an identification-dependent element u, and replace it by $a(u)$. Feasibility is preserved because $c(a(u)) \le c(u)$, and the objective does not increase; both by (H2).
(T3) Each such replacement strictly decreases the number of identification-dependent elements, because $a(u)$ is identification-free by (H2) and the status of no other element changes. The plan is finite, so the iteration terminates at an identification-free plan $\tilde{S}_\theta$ that is feasible and satisfies $j(\tilde{S}_\theta; \theta) \le j(S_\theta; \theta)$.
(T4) By θ-optimality of S_θ the reverse inequality also holds, so $j(\tilde{S}_\theta; \theta) = j(S_\theta; \theta)$.
(T5) By (H1) the value of any identification-free plan does not depend on θ. Let $j^\star$ be the minimum over identification-free plans, attained at $S^\star$; both are θ-independent.
(T6) Since $\tilde{S}_\theta$ is identification-free, $j(\tilde{S}_\theta; \theta) \ge j^\star$. Since $S^\star$ is feasible, $j^\star = j(S^\star; \theta) \ge j(S_\theta; \theta) = j(\tilde{S}_\theta; \theta)$. The two inequalities force $j(S_\theta; \theta) = j^\star$ for every θ.
(T7) Hence $\mathbb{E}_\theta[j(S_\theta; \theta)] = j^\star$.
(T8) $S^\star$ is available to the prior planner, whose optimum can only be at least as good: $\mathbb{E}_\theta[j(S_\pi; \theta)] \le j^\star$.
(T9) Substituting (T7) and (T8) into (2) gives $\mathrm{EVPI} \le 0$; with $\mathrm{EVPI} \ge 0$ this forces $\mathrm{EVPI} = 0$. □

The argument requires both hypotheses. With (H2) alone, identification-free plans are optimal but their value may still depend on θ, and step (T5) fails. The argument uses nothing specific to expected loss, which gives the following extension.

Corollary 5 (Monotone law-invariant objectives) *Let ρ be a functional of the loss distribution that is monotone with respect to almost sure dominance, and set $J_\rho(S;\theta) = \rho(L(S;\theta,\cdot))$. If (H1) holds and (H2) holds with j replaced by J_ρ, then the value of perfect structural information under J_ρ is zero. This covers expected loss, CVaR_α for every α, and spectral risk measures.*

Proof (C1) Steps (T1)–(T4) use only finiteness of the plan set, the cost clause and the dominance inequality, all assumed for J_ρ.
(C2) Step (T5) uses (H1) through the *law* of L: if the law of $L(S;\theta,\cdot)$ does not depend on θ, then neither does $\rho(L(S;\theta,\cdot))$ for any functional ρ of that law. This is where law invariance rather than mean invariance is required.
(C3) Steps (T6)–(T9) are algebraic and carry over with j replaced by J_ρ. □

5 Bounding the value outside the zero-value regime

Theorem 6 (Identification-premium bound, disjoint categories) *Suppose the focal categories' subtrees are node-disjoint, the θ_k are independent across categories, and plans decompose with $c(S) = \sum_k c(S_k)$. For a budget b define*

$$v_k^{\mathrm{inv}}(b) = \min_{c(a)\le b} \mathbb{E}_{\theta_k}\, p_k(a;\theta_k), \tag{5}$$

$$v_k^{\mathrm{full}}(b;\theta_k) = \min_{c(a)\le b} p_k(a;\theta_k), \tag{6}$$

with p_k the starvation probability of focal category k and the minima over prior-available actions. Let $\mathcal{B}_k$ be the finite set of action costs of category k not exceeding the plan budget B. Writing U for the bound, to keep it distinct from that budget,

$$\mathrm{EVPI} \;\le\; U \;:=\; \frac{1}{K_0}\sum_k I_k, \qquad I_k = \mathbb{E}_{\theta_k}\left[\max_{b\in\mathcal{B}_k}\left(v_k^{\mathrm{inv}}(b) - v_k^{\mathrm{full}}(b;\theta_k)\right)^+\right]. \tag{7}$$

Proof (S1) Every plan induces a budget split by setting $\beta_k = c(S_k)$, and every split $\{\beta_k\}$ with $\sum_k \beta_k \le B$, together with per-category actions of cost at most β_k, induces a feasible plan. The correspondence is not one-to-one, since several splits induce the same plan, but that is not needed below.
(S2) By independence of the θ_k and additivity of the loss (1), the prior objective separates across categories once the split is fixed.
(S3) Hence, with $A = \min_S \mathbb{E}_\theta j(S;\theta)$, $K_0 A = \min_{\mathrm{split}} \sum_k v_k^{\mathrm{inv}}(\beta_k)$.
(S4) Similarly, with $M(\theta) = \min_S j(S;\theta)$, $K_0 M(\theta) = \min_{\mathrm{split}} \sum_k v_k^{\mathrm{full}}(\beta_k;\theta_k)$.
(S5) Define the pointwise deficit $d_k(b;\theta_k) = v_k^{\mathrm{inv}}(b) - v_k^{\mathrm{full}}(b;\theta_k)$; it may be negative for particular θ_k, and only its positive part enters.
(S6) Both value functions are nonincreasing step functions of b whose jumps occur only at action costs, so $d_k(\cdot;\theta_k)$ is piecewise constant and $\sup_{b\le B} d_k(b;\theta_k)^+ = \max_{b\in\mathcal{B}_k} d_k(b;\theta_k)^+ =: D_k(\theta_k)$. This is what makes the premium finite to evaluate: a supremum over a continuum of budgets is attained on a finite breakpoint set.
(S7) Pointwise in θ and for every split, $\sum_k v_k^{\mathrm{full}}(\beta_k;\theta_k) \ge \sum_k v_k^{\mathrm{inv}}(\beta_k) - \sum_k D_k(\theta_k)$.
(S8) Minimising both sides over splits preserves the inequality, since $f \ge g - c$ pointwise implies $\min f \ge \min g - c$ for a constant c; the correction term does not depend on the split, which is what allows the minimisation to pass through. Hence $K_0 M(\theta) \ge K_0 A - \sum_k D_k(\theta_k)$.

(S9) Taking $\mathbb{E}_\theta$ of (S8) and using the definition of I_k in (7) gives $K_0\,\mathbb{E}[M] \geq K_0 A - \sum_k I_k$. The slack in the bound is introduced earlier, at (S6) and (S7), where a category's deficit is replaced by its maximum over budgets and the positive part discards the cases in which identification would have hurt.
(S10) Rearranging, $\mathrm{EVPI} = A - \mathbb{E}[M] \leq K_0^{-1} \sum_k I_k$. □

Corollary 7 (Dilution) *Call focal category k* isolated *if its subtree contains exactly one latent category (v, k'), the candidates in $C_{v,k'}$ have no inputs and supply no category other than (v, k'), and every other node of the subtree fails with probability zero. Let c_v be the least cost of securing a node through which v's output reaches the focal node, v included, with $c_v = \infty$ if there is none, as when $v = 0$. If focal category k is isolated, with m equal-cost, equal-risk candidates of risk $\bar{q}$, a uniform prior and securing cost c_u, then $I_k = \bar{q}(1 - 1/m)$ if $c_u \leq B$ and $c_u < c_v$, and $I_k = 0$ otherwise.*

Proof (D1) Isolation fixes the starvation probability. With supplier θ, $Z_k = F_\theta$ unless the plan secures θ or a node through which v's output reaches the focal node, in which case $Z_k = 0$: the other candidates feed nothing, and every other node of the subtree is reliable. Hence $p_k(a;\theta) = 0$ if a contains θ or such a node, and $p_k(a;\theta) = \bar{q}$ otherwise. Securing any other node changes nothing.
(D2) If $b < \min\{c_u, c_v\}$ no useful action is affordable, so both value functions equal $\bar{q}$ and the deficit is zero. If $b \geq \min\{c_v, mc_u\}$ both planners can reach zero, by the path cut or by securing every candidate, and the deficit is zero.
(D3) Suppose $c_u < c_v$ and $b \in [jc_u, (j+1)c_u) \cap [c_u, \min\{c_v, mc_u\})$ with $1 \leq j < m$. The informed planner secures the true candidate, so $v^{\mathrm{full}} = 0$. The uninformed planner can afford at most j candidates; under the uniform prior any j of them cover the supplier with probability j/m, so $v^{\mathrm{inv}} = \bar{q}(m-j)/m$. The deficit $\bar{q}(m-j)/m$ is largest at $j = 1$, on $[c_u, \min\{2c_u, c_v\})$, which is nonempty because $c_u < c_v$ and $c_u > 0$.
(D4) That interval contains the breakpoint c_u, which lies in $\mathcal{B}_k$ exactly when $c_u \leq B$. Then $D_k(\theta) = \bar{q}(1 - 1/m)$ for every θ and $I_k = \bar{q}(1 - 1/m)$. If $c_u > B$, every breakpoint in $\mathcal{B}_k$ falls under (D2). If $c_u \geq c_v$, every budget is either below $c_v \leq c_u$ or at least c_v, and again falls under (D2). In both cases $I_k = 0$. □

Isolation is not a formality. In Example 1 of Appendix C, raise the cost of securing w above c_u and keep $c_u \leq B$. Focal category 1 then has two equal-risk candidates with $\bar{q} = 0.1$, but u_1 also supplies (w, b), so the category is not isolated. Its premium is $I_1 = 0.025$, not $\bar{q}(1 - 1/m) = 0.05$: when u_2 is the supplier, securing it still leaves w exposed through u_1, and the informed planner's advantage vanishes.

When the bound is attained.

Corollary 7 gives the premium I_k exactly. It does not say that EVPI $= U$, and in general the two differ. Remark 1 gives one reason, an affordable path cut. A second is that the uninformed planner can move budget between categories. Take two isolated categories, each with $m = 2$ candidates and $c_u = 1$, risks 0.1 and 0.9, unaffordable path cuts and $B = 2$. The uninformed planner secures both candidates of the riskier category, so EVPI $= 0.05$ while $U = 0.25$, although $c_u \leq B < c_v$ in both categories. Equality does hold in the following case. Suppose the assumptions of Theorem 6 hold,

the K_0 focal categories are isolated with a common m, $\bar{q}$ and c_u and a uniform prior, every path cut costs more than B, and $K_0 c_u \le B < (K_0+1)c_u$. Then EVPI $= U = \bar{q}(1-1/m)$.
(E1) Every category has $c_u \le B < c_v$, so Corollary 7 gives $I_k = \bar{q}(1-1/m)$ for each k, and $U = \bar{q}(1-1/m)$.
(E2) Securing the K_0 true suppliers costs $K_0 c_u \le B$ and sets every Z_k to zero, so $\min_S j(S;\theta) = 0$ for every θ.
(E3) A plan the uninformed planner can afford contains no path cut and at most $\lfloor B/c_u \rfloor = K_0$ candidates, and securing any other node of the subtrees has no effect, by (D1). If it secures n_k candidates of category k, that category starves with prior probability $\bar{q}(m-n_k)^+/m$. Since $\sum_k n_k \le K_0$, K_0 times its objective is at least $\bar{q}\sum_k (m-n_k)/m \ge \bar{q}K_0(m-1)/m$, with equality at $n_k = 1$ for every k.
(E4) Hence the prior-optimal value is $\bar{q}(1-1/m)$, and by (E2) EVPI $= \bar{q}(1-1/m) = U$.

Proposition 8 (A checkable witness for (H2)) *Fix a latent category (v,k) with $v \neq 0$, so that v is itself securable, and suppose that (i) every candidate $u \in C_{v,k}$ has v as its only consumer, and (ii) $c_v \le \min_{u \in C_{v,k}} c_u$. Then securing v is an identification-free replacement satisfying both clauses of (H2) for every $u \in C_{v,k}$. Both conditions are functions of the public graph and the public cost vector, so they can be checked before any structural discovery is commissioned.*

Proof (W1) Securing v guarantees v's output to its consumers in every scenario and for every $\theta_{v,k}$, so under any plan containing that action the focal loss is a function of the scenario and of the plan's other elements alone, and its conditional law does not depend on $\theta_{v,k}$: the action is identification-free in the sense of Definition 1. The service of (v,k) itself is unchanged; what the action removes is its influence on L.
(W2) The cost clause is (ii) directly: $c_v \le c_u$ for every candidate $u \in C_{v,k}$.
(W3) For the dominance clause, fix θ and a plan S containing u. Under the conjunctive cascade, securing u affects the focal loss only through the categories u serves, and by (i) the only such category is (v,k); its effect there is to remove u's own failure from that category, which helps only when u is the true supplier and would otherwise fail. Securing v instead guarantees v's output outright, which removes the failure of (v,k) and of every other category v consumes, for every θ and every scenario.
(W4) Hence the operability of every node downstream of v under $S \setminus \{u\} \cup \{v\}$ is at least that under S, scenario by scenario, so $L(S \setminus \{u\} \cup \{v\}; \theta, \omega) \le L(S;\theta,\omega)$ pointwise, and therefore $j(S \setminus \{u\} \cup \{v\};\theta) \le j(S;\theta)$ for every θ. With (W1) and (W2), (H2) holds with $a(u) = v$. □

The hypothesis $v \neq 0$ is not cosmetic. Plans are subsets of $V \setminus \{0\}$, so the focal node cannot be secured and the witness is unavailable for exactly the K_0 categories that enter the loss directly. Proposition 8 closes the gap left by (H2) in one direction only. It gives a sufficient condition a firm can test on its own graph and cost data; it does not decide (H2) in general, and its condition (i) fails wherever a candidate supplies more than one consumer. Section 8 records what remains open.

Remark 1 (The premium is loose where the theorem bites) The maximum in (7) ranges over all breakpoints, including budgets a planner would not choose. If $c_v \le B$, so that the consumer cut is affordable, both planners take it and reach zero loss for that category, so

its contribution to EVPI is zero; this follows directly, not through Theorem 4, whose cost clause does not hold when $c_u < c_v$. Yet Corollary 7 still returns $I_k = \bar{q}(1 - 1/m) > 0$ in exactly that case. The bound remains valid, being an upper bound, but it is not tight there, and a screening procedure built on it can flag channels whose decision value is zero. This accounts for a substantial part of the mass near zero in Figure 1: among the enumerated generic instances in which every category has an affordable cut, the mean tightness ratio is 0.102 and 67% of instances fall below 0.05, against 0.203 and 41% among those in which no category does. The looseness is a property of the bound itself.

Corollary 9 (Per-channel bound) *Under the assumptions of Theorem 6, the partial value of learning θ_k alone satisfies* $\mathrm{EVPPI}_k \le I_k/K_0$, *where* $\mathrm{EVPPI}_k = A - \mathbb{E}_{\theta_k}\big[\min_S \mathbb{E}_{\theta_{-k}} j(S;\theta)\big]$.

Proof (P1) With θ_k known and the remaining components unknown, the separation of step (S2) gives, for each split, $K_0\,\mathbb{E}_{\theta_{-k}} j = v_k^{\mathrm{full}}(\beta_k;\theta_k) + \sum_{j\neq k} v_j^{\mathrm{inv}}(\beta_j)$.
(P2) By (S6), $v_k^{\mathrm{full}}(\beta_k;\theta_k) \ge v_k^{\mathrm{inv}}(\beta_k) - D_k(\theta_k)$ for every β_k.
(P3) Adding $\sum_{j\neq k} v_j^{\mathrm{inv}}(\beta_j)$ to both sides of (P2) and minimising over splits, permissible because $D_k(\theta_k)$ does not depend on the split, gives $K_0 \min_S \mathbb{E}_{\theta_{-k}} j \ge K_0 A - D_k(\theta_k)$.
(P4) Taking $\mathbb{E}_{\theta_k}$ and rearranging gives $\mathrm{EVPPI}_k \le I_k/K_0$. □

5.1 Dispersion under a specified construction

The empirical study introduces heterogeneous candidate risk governed by a parameter λ. Monotonicity in a vague notion of "more dispersion" would not be well posed, since it leaves open which ordering of distributions is meant, so we fix the family and compute the premium exactly.

Definition 2 (Two-point dispersion family $\mathcal{D}(\lambda)$) A latent category has m candidates and a uniform prior over which of them is the supplier. Candidate risks take the two values $\bar{q} \pm \lambda\delta$ with equal probability, independently of which candidate is the supplier, where $\lambda \in [0,1]$, $\delta > 0$ and $\bar{q} \pm \delta \in [0,1]$. The assignment of risks to candidates is latent and exchangeable under the prior, and is revealed only by probing. The category is an isolated focal category in the sense of Corollary 7, and c_v is the least cost of a path cut as defined there. Actions are: no action at cost 0; secure a named candidate at cost c_u; secure the consumer at cost $c_v > c_u$.

Proposition 10 (Premium under $\mathcal{D}(\lambda)$) *Write $D = \bar{q}(1 - 1/m)$ for the dilution term of Corollary 7. Under $\mathcal{D}(\lambda)$,*

$$I_k(\lambda) = \begin{cases} \tfrac{1}{2}\max\{\lambda\delta,\, D\} + \tfrac{1}{2}D, & c_u \le B, \\ \tfrac{1}{2}\lambda\delta, & c_u > B. \end{cases} \tag{8}$$

In both cases I_k is nondecreasing in λ. In the first it is constant for $\lambda \le D/\delta$ and strictly increasing above that threshold; in the second it is strictly increasing throughout. If the risk assignment is instead public, the prior planner can target the fragile candidate directly and the premium need not increase in λ.

Proof (R1) At $b = 0$ the only available action is inaction, so $v^{\text{inv}}(0) = \mathbb{E}[q_\theta] = \bar{q}$ by the mean-preserving construction, while $v^{\text{full}}(0;\theta) = q_\theta$. The deficit is $\bar{q} - q_\theta$, whose positive part equals $\lambda\delta$ when the supplier carries the low risk and 0 otherwise, each with probability $\frac{1}{2}$.
(R2) For $b \in [c_u, c_v)$ the informed planner secures the supplier, so $v^{\text{full}} = 0$. The uninformed planner faces exchangeable candidates — this is where latency of the assignment is used — so every named candidate has the same prior value; securing one leaves the category exposed with probability $(m-1)/m$ at conditional risk $\bar{q}$, giving $v^{\text{inv}} = D$. This is below the do-nothing value $\bar{q}$, so it is the prior planner's best action on that interval. The deficit is D for both risk types.
(R3) For $b \geq c_v$ both planners cut the consumer and reach zero, so the deficit is zero.
(R4) If $c_u \leq B$, the maximum over breakpoints is $\max\{\lambda\delta, D\}$ when the supplier carries the low risk, and D when it carries the high risk. Averaging over the two equally likely types gives the first line of (8).
(R5) If $c_u > B$, only the breakpoint $b = 0$ is feasible and (R1) gives the second line.
(R6) Both expressions are nondecreasing in λ because $\lambda \mapsto \max\{\lambda\delta, D\}$ and $\lambda \mapsto \lambda\delta$ are nondecreasing; the threshold statement follows from $\lambda\delta \geq D \iff \lambda \geq D/\delta$.
(R7) If the assignment is public, the prior planner may secure the high-risk candidate deliberately, which changes v^{inv} and removes the exchangeability used in (R2), so monotonicity no longer follows from this argument. □

Equation (8) predicts a threshold rather than a slope. Below $\lambda = D/\delta$ the dilution term dominates and dispersion contributes nothing to the premium; only above it does dispersion matter. Section 7.5 reports what the implemented planner does over the same range.

5.2 Why the bound does not transfer to CVaR

Theorem 4 extends to CVaR at no cost (Corollary 5). Theorem 6 does not, and the obstruction is structural.

Proposition 11 (What can be said) *Let $L \in [0,1]$. Then $\mathbb{E}[L] \leq \text{CVaR}_\alpha(L) \leq \mathbb{E}[L]/(1-\alpha)$, and, writing $\text{EVPI}_{\text{CVaR}}$ for the value of perfect information under the CVaR objective,*

$$\text{EVPI}_{\text{CVaR}} \;\leq\; \frac{1}{1-\alpha}\,\text{EVPI} \;+\; \frac{\alpha}{1-\alpha}\,\mathbb{E}_\theta\Big[\min_S j(S;\theta)\Big]. \tag{9}$$

Proof (V1) By the Rockafellar–Uryasev representation [23], $\text{CVaR}_\alpha(L) = \min_t\{t + (1-\alpha)^{-1}\mathbb{E}(L-t)^+\}$.
(V2) Evaluating at $t = 0$ and using $L \geq 0$ gives $\text{CVaR}_\alpha(L) \leq \mathbb{E}[L]/(1-\alpha)$.
(V3) For any t, since $(1-\alpha)^{-1} \geq 1$ and $\mathbb{E}(L-t)^+ \geq 0$, $t + (1-\alpha)^{-1}\mathbb{E}(L-t)^+ \geq t + \mathbb{E}(L-t)^+ \geq t + \mathbb{E}(L-t) = \mathbb{E}[L]$; minimising over t gives $\text{CVaR}_\alpha(L) \geq \mathbb{E}[L]$.
(V4) With $A_{\text{CVaR}} = \min_S \mathbb{E}_\theta J(S;\theta)$, (V2) gives $A_{\text{CVaR}} \leq (1-\alpha)^{-1}A$.
(V5) By (V3), $\mathbb{E}_\theta \min_S J(S;\theta) \geq \mathbb{E}_\theta \min_S j(S;\theta) = \mathbb{E}[M]$.
(V6) Subtracting and using $A = \text{EVPI} + \mathbb{E}[M]$ yields (9). □

Inequality (9) is not a multiplicative transfer. Its second term is proportional to the residual loss $\mathbb{E}_\theta[\min_S j(S;\theta)]$, the expected loss that remains after optimal mitigation, which is of the same order as the objective itself. At $\alpha = 0.9$ the coefficient $\alpha/(1-\alpha)$

is 9, so that term is far larger than any advantage measured here. We therefore do not use the inequality. The obstruction is visible at step (S2) of Theorem 6, which requires the objective to be *additive* across categories. CVaR is subadditive, and subadditivity gives an inequality in the direction that destroys the decomposition. A CVaR version of Theorem 6 would need a different argument, which we do not have.

5.3 Role of CVaR in the empirical model

Since the premium bound is stated for expected loss, it is reasonable to ask why the empirical planner optimises CVaR. Corollary 5 makes the zero-value result a statement about risk-sensitive planning as well, and we preferred to take the measurements under a risk-sensitive objective rather than under the one for which the result is easiest to obtain. Theorem 6 is not used to bound any CVaR measurement reported in Section 7. One consequence is that the only actionable artefact the theory produces, the bound U, is never on the same objective as any measurement we report. Repeating the benchmark measurements under expected loss is a configuration change rather than a new experiment, and it would place the ceiling of Section 7.7 and the advantage of Section 7.3 on one objective. We have not done it.

5.4 Unresolved extension: shared subtrees

When category subtrees share nodes, step (S2) fails and the decomposition fails with it. We state what we believe and have not proved.

Conjecture 12 *Under a single-cut regime, meaning that optimal plans take at most one action per focal-category channel, and with replacement slack, the bound* (7) *holds when each shared channel's premium is counted once per focal category it feeds.*

Section 6 reports an enumeration search for counterexamples that found none, which is evidence about a finite class of small instances and not a proof.

Table 3 collects the assumptions each result needs and where it applies, since the hypotheses differ from result to result and the distinctions matter for what may be claimed of the measurements.

6 Computational verification

The experiments in this section are not offered as proof of the analytical results. They serve two purposes: verifying that the implementation matches the definitions, and searching finite instances for violations that would reveal errors in the stated results. In each category a plan may take no action, cut at the consumer, or secure any nonempty subset of the candidates. Every instance is enumerated exactly, over all plans and all θ, with no sampling inside an instance; seeds are used only to draw the instances themselves. The pipeline is deterministic and reproduced identical reported outputs on an independent machine.

Table 4 reports the three batches, and Figure 1 shows the distribution of the ratio of exact EVPI to the bound. Most instances lie near zero, so the bound is loose in

Table 3 Assumptions, results and where each applies.

result	required assumptions	establishes	applicability
Lem. 2	finite feasible plan set	acquisition value $\leq$ EVPI	general
Prop. 1	implemented planner defined in both regimes	$H = \text{EVPI} + \varepsilon_\pi - \varepsilon_\theta$	general
Thm. 4	(H1), (H2)	$\text{EVPI} = 0$	(H1) via (H1′), checkable in advance; (H2) via Prop. 8 or a plan-space argument
Prop. 8	$v \neq 0$, candidates sole-consumed by v, $c_v \leq \min_u c_u$	(H2) holds for that category	checkable from public data; holds on 2.8% of latent categories here
Cor. 9	as Thm. 6	$\text{EVPPI}_k \leq I_k / K_0$	per category
Cor. 5	(H1), (H2) for a monotone law-invariant objective	zero value under CVaR	theoretical
Thm. 6	disjoint subtrees, independent θ_k, additive budget	$\text{EVPI} \leq U$	subset of networks
Cor. 7	isolated category, equal cost and risk, uniform prior	exact premium I_k; EVPI $= U$ only in a sub-case	special case
Prop. 10	family $\mathcal{D}(\lambda)$	premium in closed form	the overlay used here
Conj. 12	shared channels, single cut, slack	unproved bound	exploratory
SISS	additional heuristic construction	screening score	not validated

Table 4 Enumeration checks. A single violation would falsify the corresponding statement.

batch	inst.	viol.	auxiliary check	tightness
generic random	6000	0	—	mean 0.132, max 0.985
Thm. 4 regime	2000	0	nonzero-EVPI cases: 0	—
Cor. 7 regime	2000	0	formula mismatches: 0	mean 0.212, max 1.000

general. The dilution batch is loose too, with mean ratio 0.212: the premium is exact there, but the bound is not. Equality is reached only on some of its instances, and it holds exactly in the sub-case identified after Corollary 7, so the bound cannot be multiplied by any constant smaller than one without further assumptions.

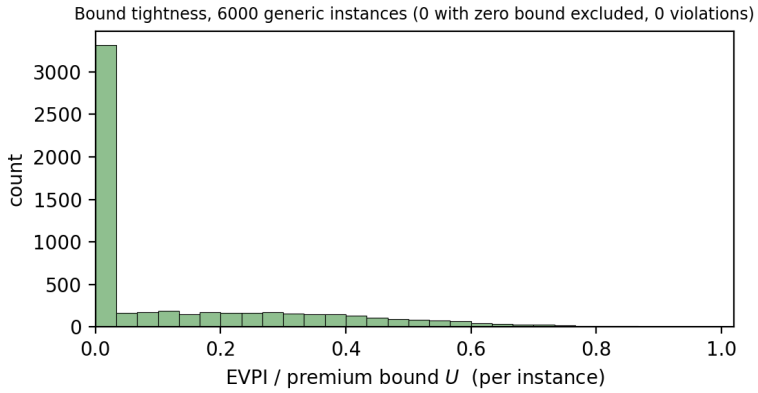


Fig. 1 Ratio of exact EVPI to the bound of Theorem 6 over 6000 enumerated instances. No value exceeds one.

The premium against exact EVPPI.

On miniature disjoint instances we computed the exact per-channel EVPPI_k by enumeration and compared it with the quantity Corollary 9 says bounds it, namely I_k/K_0. Both sides carry the $1/K_0$ factor, so the slack reported here is slack in the bound and not the divisor. Over five independent seeds and 24,981 channels, I_k/K_0 never fell below the enumerated EVPPI_k, and the whole-instance bound was never violated. Rank agreement is moderate and stable across seeds (Spearman 0.505 to 0.516; Pearson 0.597 to 0.615), with the bound typically about 2.7 times the quantity it bounds.

Search for a counterexample to Conjecture 12.

Over five seeds and 100,000 instances in which channels feed several focal categories, the conjectured bound was never violated and the minimum slack was zero, so it is attained with equality on some instances. The search covers instances with at most three channels and three focal categories; a counterexample may exist outside that range.

7 Empirical study

7.1 What is real and what is modelled

The topologies are real. The Willems dataset [30] is a public INFORMS/MSOM supplement of multi-echelon supply chain structures. Everything else is modelled: which candidate is the true supplier, the risk attributes candidates carry, the probe channel, the disruption correlation structure and the failure probabilities are all generated by the model of Section 3. No public dataset records observed sub-tier supplier identities, attributes or probe outcomes. We therefore refer throughout to benchmark network topologies with model-generated latent overlays, and no conclusion rests on observed disruption outcomes.

Focal categories.

How K_0 is determined depends on the chain. Where a chain has several sinks, the virtual market focal consumes one category per end product, so K_0 is the number of sinks; where a chain has a single sink, that sink is itself the focal node and K_0 is the number of input categories it consumes. Across the 32 retained chains, every one of which the loader reports as multi-sink, K_0 ranges from 2 to 672 with a median of 52. The comparison with the values $K \in \{2, 3, 4\}$ at which exact enumeration is possible in Section 7.8 is therefore not uniform: most chains lie far above that range, but three of the 32 have $K_0 = 2$, inside it. At $K = 4$, the largest enumerable value, H already understates EVPI on 63% of instances, and how the discrepancy behaves at 52 or 672 is not known. That is why the trend of Section 7.8 is not extrapolated to the benchmark measurements, in either direction.

One consequence of the construction should be stated here rather than left to be inferred. The virtual market focal consumes one category per end product and each end product leaves the chain through exactly one sink, so every focal category is served by a single candidate by construction, as the census of Section 7.3 confirms. The tier-one layer, the one nearest the loss, is therefore fully known in every instance, and structural uncertainty can reach the loss only indirectly through a cascade from tier two or deeper. A design in which the nearest layer is transparent should be expected to measure a small advantage; that is a property of the construction and not a finding about supply networks.

The two regimes.

The *public-attribute* regime is $\lambda = 0$: every candidate's risk attribute is common knowledge, and only the supplier identities are latent. The *latent-attribute* regime is $\lambda = 0.8$: nodes at tier two or deeper additionally carry a mean-preserving two-point risk attribute, drawn with dispersion λ about the node's nominal rate, whose assignment to candidates is hidden until probed. This overlay is the network analogue of the family $\mathcal{D}(\lambda)$ of Definition 2, not an instance of it: it disperses at node level across a cascade rather than within one isolated category with the three actions of Definition 2. Comparisons between Proposition 10 and the measurements are therefore qualitative, and we treat them as such in Section 7.5.

The synthetic family.

Twelve networks drawn from the scale-free family of the pipeline's generator, at a target size of 1000 nodes. The generator allocates that total across tiers and rounds each tier to an integer, which yields realised sizes of about 977; correlation regions are assigned by inheritance from a node's consumer, so a supplier usually shares its consumer's region. Unlike the benchmark topologies, which are real and carry a modelled overlay, these networks are generated in full. The generator and its parameters accompany the code. Twelve is the number of evaluation networks the pipeline draws per family; it is too few for population claims, which is why we treat this family as a set of structural stress tests throughout.

7.2 Design and preprocessing

Table 5 Principal parameters of the measured problem. The node cost model, the nominal failure rates, the candidate pool sizes and the copula correlation are set in the configuration accompanying the code. The tail level α, the scenario counts and the mitigation budget B all govern the numbers in Sections 7.3–7.7; B in particular fixes which actions are affordable and so which breakpoints exist in (7).

parameter	value	source
tail level α	0.9	configuration
mitigation budget B	9.0	run report, all reported runs
scenario draws per evaluation	160	configuration
scenario batch used in planning	200	configuration
correlation clusters	6 regions	configuration default
seeded evaluations per instance	10 / 16 / 8	Tables 7, 8, 9

Table 5 lists the parameters of the measured problem. A cost-budgeted greedy backup-qualification planner is held identical across all conditions; candidate pools are built only from prior and public information; the prior over latent structure is sampled and calibrated to the public sourcing multiplicity marginal, nothing being conditioned on any observation; evaluation is paired by seed. Planning and evaluation use disjoint scenario batches (Table 5), so every advantage reported below is measured out of sample with respect to the scenarios the planner optimised against; a near-zero advantage therefore cannot be explained by the planner having overfitted its own batch. A leakage audit confirms that the uninformed planner's output does not change when the hidden ground truth is permuted. The wider pipeline also implements a probe channel, but neither EVPI nor H involves probing: the uninformed planner sees only the prior and the informed planner sees θ outright, so no probe parameter enters any number reported below.

Table 6 Preprocessing of the benchmark collection.

step	networks	criterion
released collection	38	Willems [30]
minimum size filter	−6	fewer than 30 nodes in the focal cone. The six excluded chains are 01–06, with cones of 9, 14, 18, 23, 28 and 29 nodes; such chains have too few latent tiers for discovery to have anything to find
retained	32	each given a virtual market focal consuming every end product as its own category when the chain has several sinks
evaluated instances	96	three seeded overlays per retained network

The minimum-size threshold in Table 6 is 30 nodes in the focal cone. It is the pipeline default and was applied before any outcome was observed; the retained set is chains 07–38, which is what the archived result files contain. No chain lies near the threshold: the largest excluded cone has 29 nodes and the smallest retained one has 39, so the partition does not depend on the exact value chosen or on whether the virtual market focal is counted. The rule is not neutral, however: it removes the smallest chains and biases the retained set towards deeper, larger topologies. Since Section 7.6 discusses which networks show positive value, that bias should be kept in mind; the excluded chains would, if anything, be expected to show less value rather than more.

The 96 instances comprise three overlays per underlying network and are therefore not independent network replicates. Summary statistics are computed at instance level; reliability is assessed at network level; and proportions over instances should be read as clustered within 32 networks.

7.3 Public-attribute regime

Table 7 Heuristic oracle advantage H in the public-attribute regime, $\lambda = 0$.

dataset	inst.	mean	median	exact zeros	> 0.01	frac > 0.02
Benchmark topologies	96	+0.0011	0.0000	24	3	0.010
Synthetic $n \approx 977$	12	+0.0028	0.0000	5	3	0.167

Six of the 96 benchmark instances have an advantage exceeding twice its own standard error, computed across the seeded evaluations of that instance, and the largest single instance reaches 0.0212. That count is a resolution statement and should be read as one: with 160 evaluation scenarios at $\alpha = 0.9$ each CVaR estimate averages the worst 16 draws, so 90 of the 96 instances are not separable from zero at that criterion. The count of six is not easy to read either way: a one-sided two-standard-error rule applied to 96 draws from a true null flags about 2.2 on average, and a Poisson approximation puts $P(X \geq 6)$ at about 0.025, so six is mildly inconsistent with a global null while remaining consistent with the advantage being resolvable on only a handful of instances. The mean is better resolved than any single instance, but not by as much as an unclustered calculation suggests, and not enough to call it separated from zero. Taking the 96 instances as independent gives a standard error of 0.00038 against a mean of 0.00106, a ratio of 2.8, and 2.2 at $\lambda = 0.8$. Overlays are clustered within chains (Section 7.2), however, and computed over the 32 chain means the ratio falls to 1.92 at $\lambda = 0$ and, over the 14 chains, to 1.96 at $\lambda = 0.8$, two-sided $p = 0.064$ and $p = 0.072$. The clustered figures are the ones to read, and on them neither mean is significantly different from zero at conventional levels. We therefore report means and medians as descriptions of the distribution rather than as measurements separated from zero, and the comparisons of Section 7.5 inherit that limitation. A quarter of the instances give exactly zero. Figure 2 shows the distributions. We offer no explanation of the mass at zero from Theorem 4. At $\lambda = 0$ the construction makes candidates

within a category approximately exchangeable in marginal law, which is not sufficient for (H1) (Section 4); and even (H1) alone permits positive EVPI, since the θ-optimal plan may be identification-dependent and strictly better than every identification-free plan, so consistency with (H1) predicts nothing about the mass at zero. Nor is (H2) enforced by construction: the pipeline prices every securing action by a single cost attached to the node secured, so whether an identification-free replacement is no dearer than the identification-dependent action it replaces is decided by the two node costs and holds for some categories and not others. We can, however, say how far the checkable route of Proposition 8 goes on these topologies, since both of its conditions are read off the public graph and cost vector. Across the 32 retained chains there are 14,977 categories, of which 13,313 (88.9%) have a candidate pool of size one and so carry nothing to identify: their contribution to the value of structural information is zero by construction, not by any theorem. Of the 1,664 genuinely latent categories, 106 (6.4%) satisfy the sole-consumer condition, 210 (12.6%) satisfy the cost condition, and 47 (2.8%) satisfy both, so the witness holds on 2.8% of them. The first group provably contributes nothing: with one candidate there is no θ to learn. The second does not. Proposition 8 establishes (H2) for a category, while Theorem 4 is a global statement that also requires (H1), and no per-category zero-value result appears anywhere in this paper. The 47 are categories in which one of the two hypotheses is verifiably met, and nothing more. None of them sits at the focal node, as the hypothesis $v \neq 0$ of Proposition 8 requires: every focal category in these chains has a candidate pool of size one. That is forced by the virtual market construction rather than observed (Section 7.1), which also makes the hypothesis $v \neq 0$ close to a formality here: the categories the witness cannot reach are exactly the categories with nothing to identify.

Two things follow, and the second is unwelcome. Proposition 8 does not explain the mass at zero: at 2.8% of latent categories it is too rare to be the mechanism, and the sole-consumer condition is what binds, as one would expect on chains with shared suppliers. What does explain a large part of the observed concentration is candidate multiplicity, or its absence: nearly nine categories in ten in this collection have one possible supplier. That is a property of the benchmark collection, not a result about supply networks, and it should temper how far the measurements of this section are read as evidence about networks in general. What determines the residual mass at zero among the latent categories is unidentified, as Section 8 concedes, and the agreement with Theorem 4 is not evidence that its conditions hold.

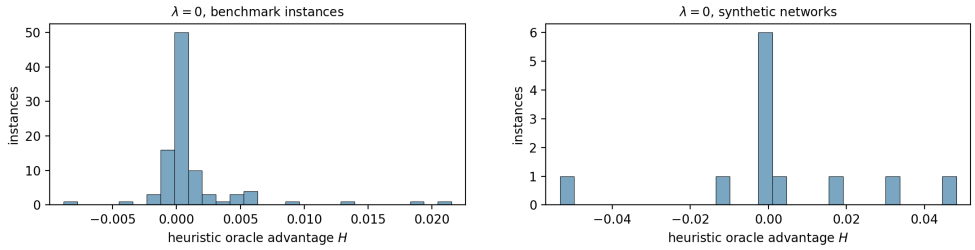


Fig. 2 Heuristic oracle advantage at $\lambda = 0$: benchmark instances (left) and synthetic networks (right).

7.4 Latent-attribute regime

Table 8 Heuristic oracle advantage at $\lambda = 0.8$. The benchmark row uses the 14 chains with at least four tiers and at most 350 nodes (7, 8, 9, 11, 12, 14, 15, 16, 17, 18, 19, 20, 21, 24), three overlays each, 16 seeded evaluations per instance; the deeper, smaller chains are those in which latent structure has something to hide. Table 7 covers all 32 retained chains at 10 seeded evaluations per instance, and the sweep of Section 7.5 uses 8. All three counts are taken from the archived result files. Seeds are nested within instances, and the instance is the unit of analysis throughout (Section 3.2).

dataset	inst.	median	mean	frac > 0.02	max
Benchmark, 14 chains	42	0.0004	0.0048	0.048	0.0915
Synthetic $n \approx 977$	12	0.0033	0.0038	0.167	0.0438

The two regimes differ in these ways: the benchmark median is 0.0004 here against exactly zero in Table 7, the fraction above 0.02 is 0.048 against 0.010, and the largest single instance is more than four times as large. The instance sets also differ, so nothing here isolates the effect of λ; Section 7.5 puts that question on a fixed set and finds no monotone relationship, with the highest values of both summaries at $\lambda = 0$. The levels remain small under the normalisation of Section 3. The synthetic family comprises 12 networks and cannot support population claims; we treat it as a set of structural stress tests and the benchmark collection as the empirical anchor. The figure of 0.167 is two networks out of twelve and should not be read as a rate.

Figure 3 shows the instance-level spread behind these summaries.

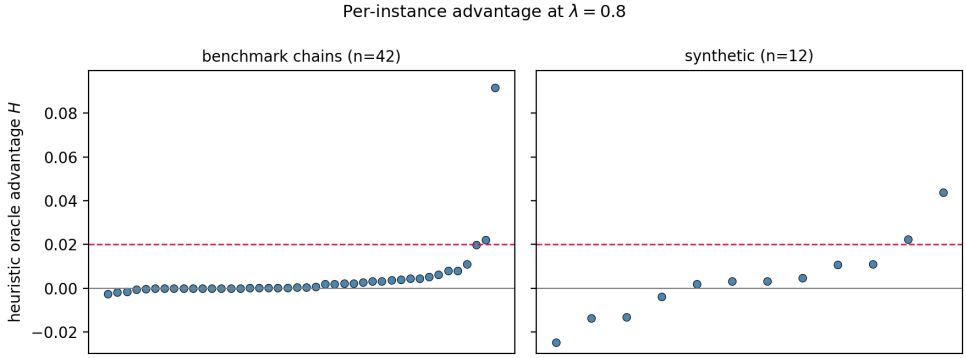


Fig. 3 Per-instance advantage at $\lambda = 0.8$. The dashed line marks 0.02, the level used in the fraction columns of Tables 8 and 9.

7.5 Dispersion on a fixed instance set

Tables 7 and 8 differ in λ and also in which instances are measured, so they cannot be compared directly. We therefore fixed eight benchmark chains (7, 9, 11, 14, 17, 18, 19, 21; three overlays each, 24 instances, 8 seeded evaluations per instance) and swept λ with everything else held constant within the sweep. The registered summary, fixed before the sweep because the median is exactly zero on most of these chains, is the mean advantage.

Overlay seeding moves these numbers more than the treatment does: at $\lambda = 0$ the sweep's maximum over eight chains is 0.0557 while the 32-chain run of Table 7 gives 0.0212, a factor of 2.6 on the same topologies under a different overlay draw. No statement about levels across tables is interpretable, and none is made. The sweep is a separate execution with its own overlay seeding, so its $\lambda = 0$ and $\lambda = 0.8$ columns are *not* the corresponding columns of Tables 7 and 8 restricted to these eight chains; they are different draws of the latent structure on the same topologies, at a different seed count. The instance-level numbers therefore differ, and the maxima in Table 9 are not comparable with those in Tables 7 and 8. Table 9 is internally comparable across its own columns, which is what the question about λ requires, and nothing outside this subsection is inferred from it.

Table 9 Dispersion sweep on a fixed set of 24 benchmark instances.

λ	0.0	0.2	0.4	0.6	0.8	1.0
mean H	0.0063	0.0020	0.0027	0.0047	0.0045	0.0031
median	0.0000	0.0000	0.0000	0.0008	0.0001	0.0000
fraction > 0.02	0.125	0.042	0.042	0.042	0.042	0.042
maximum	0.0557	0.0260	0.0549	0.0733	0.0585	0.0881

The registered statistic, the Spearman correlation of λ with the mean advantage, is -0.086 with $p = 0.87$. That test is close to powerless: on six grid points a rank correlation must reach $|\rho| \geq 0.886$ to attain $p < 0.05$, so it could only have detected a near-perfect monotone trend. It also discards the pairing, since the same 24 instances are measured at every λ. We therefore also ran the within-instance analysis over the 144 paired observations, as a stated post-hoc. Relative to $\lambda = 0$ the mean paired difference is -0.0043 at $\lambda = 0.2$ (paired t, $p = 0.028$; Wilcoxon $p = 0.108$) and -0.0036 at $\lambda = 0.4$ ($p = 0.021$; 0.021), and is not distinguishable from zero at 0.6, 0.8 and 1.0. Every difference is negative, but that is one fact rather than five: all five contrasts share the $\lambda = 0$ cell, which is the highest in Table 9 on both the mean and the fraction above 0.02, so a high baseline draw sends all of them the same way. The analysis that does not privilege a baseline is a within-instance regression of H on λ across all 144 observations, which we have not run. Holm correction across the five comparisons leaves none significant, the smallest adjusted p being 0.105, so we claim no effect. What the paired analysis does establish is that the data give no support to an increase in the advantage with λ, which the six-point rank test was too weak to have shown either way.

Nothing in the theory requires otherwise. Proposition 10 concerns the premium, an upper bound, and it is stated for the family $\mathcal{D}(\lambda)$; the network overlay is an analogue of that family rather than an instance of it (Section 7.1), so its threshold D/δ cannot be evaluated here and we do not claim the measurements illustrate it. Even where the premium does rise, Proposition 1 shows a rising bound need not produce a rising H, since H differs from EVPI by two planner optimality gaps of undetermined relative size. Neither argument accounts for the dip from 0.0063 at $\lambda = 0$ to 0.0020 at $\lambda = 0.2$, which we leave unexplained. Figure 4 places the two panels on a common horizontal axis. The maximum advantage rises with λ (Spearman +0.771, $p = 0.072$); this is recorded as exploratory, since the maximum was not the registered summary.

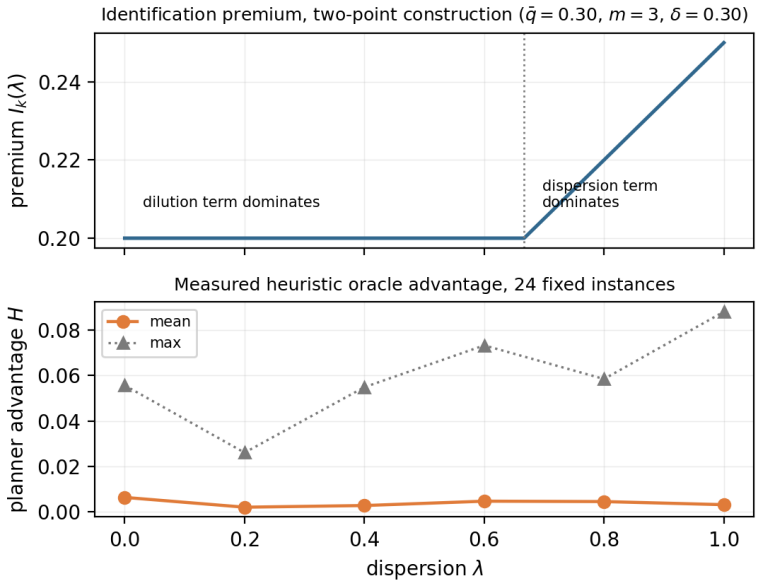


Fig. 4 Above: the identification premium of Proposition 10 at illustrative parameters, showing the threshold below which dispersion does not enter the bound. Below: the measured advantage on the fixed instance set. The upper panel is a bound on EVPI and the lower a planner-specific measurement; Proposition 1 explains why they need not move together. The vertical scales differ.

7.6 Which networks show positive value

Figure 5 shows the per-chain picture at $\lambda = 0.8$: one chain carries most of the mass and the rest sit close to zero. Three networks appear at the top in both regimes, one benchmark chain and two synthetic networks. Benchmark chain 09 has a mean advantage of +0.015 at $\lambda = 0$, from overlays of +0.0212, +0.0192 and +0.0053, and 0.034 at $\lambda = 0.8$ with a maximum overlay of 0.0915. Synthetic networks 03 and 07 are in the top three at both levels. Persistence is not universal: synthetic network 06 has the largest advantage at $\lambda = 0$ (+0.047) and almost closes at $\lambda = 0.8$ (+0.005), while

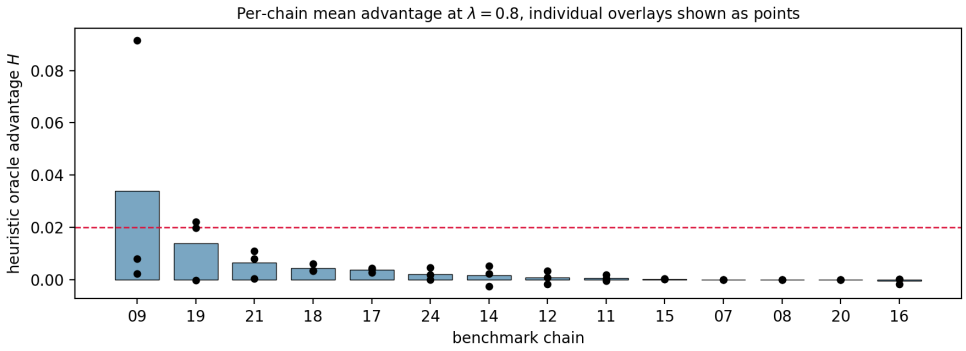


Fig. 5 Per-chain mean advantage at $\lambda = 0.8$ over the 14 benchmark chains of Table 8, with the three overlays of each chain shown as points. The dashed line marks 0.02. One chain accounts for most of the mass, and its three overlays differ by more than an order of magnitude.

networks 02 and 08 are exactly zero at $\lambda = 0$ and open at $\lambda = 0.8$. We could not identify what distinguishes them. The correlations below are computed at instance level, one observation per network–overlay pair. Since overlays are clustered within chains, the effective number of independent units is the number of chains, so the reported p-values overstate precision; none of them supports a claim in either direction, and the point of reporting them is that they do not. We write *chokepoint census* for the number of categories at tier two or deeper whose candidate pool has size one, so that the category fails whenever that single candidate does. These are exactly the categories that carry no identification problem in the census of Section 7.3; the census counts them as structural single points of failure rather than as objects of learning.

A fourth predictor is available and was not part of the pre-specified set: K_0 itself, which enters the loss as the normalising constant. It is also a chain-level covariate, constant across the three overlays of a chain, so an instance-level correlation triples the apparent sample without adding any information about K_0; the chain-level column of Table 10 is the only one that means anything for it. At chain level the scaled advantage $H \cdot K_0$ rises with K_0 at $\lambda = 0.8$ with $\rho = +0.303$ and $p = 0.293$, against $\rho = +0.349$ and $p = 0.024$ for the same comparison computed over instances: the apparent significance was an artefact of treating three overlays of a chain as three observations of K_0. Read directly, the sign would say that larger chains have more focal categories rescued in the latent-attribute regime while showing a similar normalised advantage, so part of what is reported there as a small number would be the $1/K_0$ normalisation rather than a small effect. We do not claim it: the comparison is post-hoc, uncorrected, and does not survive clustering.

The advantage itself is a different matter. At $\lambda = 0$ and at chain level it is related to K_0 with $\rho = -0.329$, $p = 0.066$, which is indistinguishable from the chokepoint census result in the row above ($\rho = -0.331$, $p = 0.065$). The two predictors are close to collinear on this collection: the census counts categories whose candidate pool has size one, and both grow with the number of end products a chain carries. What Table 10 shows at $\lambda = 0$ is therefore best read as one marginal association measured twice, not two, and it points the same way as the census result discussed below.

Table 10 Rank correlations of the advantage with the three pre-specified predictors, and below them the two post-hoc predictors of the text, at instance level and at chain level. Overlays are clustered within chains, so the chain-level column is the one with independent units; the instance-level column is shown because it is what an unclustered analysis would report. K_0 is constant within a chain, so only its chain-level row carries information.

		instance level		chain level	
regime	predictor	ρ	p	ρ	p
$\lambda = 0$	chokepoint census	-0.176	0.086	-0.331	0.065
$\lambda = 0$	network size	-0.044	0.671	-0.147	0.421
$\lambda = 0$	tier depth	-0.074	0.476	-0.183	0.315
$\lambda = 0.8$	chokepoint census	$+0.235$	0.135	$+0.161$	0.582
$\lambda = 0.8$	network size	$+0.269$	0.085	$+0.351$	0.218
$\lambda = 0.8$	tier depth	$+0.020$	0.902	$+0.070$	0.813
$\lambda = 0$	K_0	-0.157	0.130	-0.329	0.066
$\lambda = 0$	$H \cdot K_0$	-0.079	0.440	-0.258	0.154
$\lambda = 0.8$	K_0	$+0.218$	0.170	$+0.139$	0.635
$\lambda = 0.8$	$H \cdot K_0$	$+0.349$	0.024	$+0.303$	0.293

Table 10 reports the three pre-specified predictors at both levels, with the two post-hoc predictors below them. None of the pre-specified three reaches significance at either level, and the chain-level column, the one with independent units, is uniformly weaker for the two predictors that look promising at $\lambda = 0.8$. The strongest chain-level association runs opposite to the registered prediction: at $\lambda = 0$ the chokepoint census is *negatively* related to the advantage ($\rho = -0.331$, $p = 0.065$ over 32 chains), where the pre-specified hypothesis predicted a positive one. The sign is not a surprise under the account of Section 7.3. As defined here the census counts categories with a candidate pool of size one, which are exactly the categories with nothing to identify, so a chain with more of them has less latent structure and less for perfect information to act on. The registered prediction was framed when the census was read as a measure of fragility rather than of single-sourcing. With six tests and no correction we do not treat the association as a finding, and the determinant of the positive-value instances remains open.

7.7 Expressing the measurements in money

Let V be the value at risk across the focal basket over the exposure window the disruption model represents. Two monetary quantities can be formed, and they must be kept apart. They differ in two ways, not one. The first is planner-specific and the second is not; and the first is a CVaR quantity while the second is stated for expected loss, which matters because Proposition 11 shows that U does not bound the CVaR value of perfect information.

The first is planner-specific. If H is the advantage measured for an implemented planner, $H \cdot V$ is the value that perfect structural information delivered to *that* planner.

By Proposition 1 it is not an upper bound on what an exact optimiser could achieve. Table 11 reports it as a percentage of V, so that no currency assumption of ours enters.

Table 11 Planner-specific value of perfect structural information, as a percentage of the value at risk V.

regime	median	90th percentile	maximum
Benchmark, $\lambda = 0$ (96 instances)	0.00%	0.42%	2.12%
Benchmark, $\lambda = 0.8$ (42 instances)	0.04%	0.81%	9.15%

The second is a genuine ceiling. Under the assumptions of Theorem 6, $U \cdot V$ is a ceiling on the exactly-optimised value of perfect structural information, so a discovery programme priced above it cannot pay for itself against an optimally planned baseline. It is not a ceiling on H: by Proposition 1 a heuristic planner's advantage exceeds EVPI whenever $\varepsilon_\pi > \varepsilon_\theta$, which happens on 14.8% of the enumerated instances of Table 12. Computing U for a specific firm means evaluating the premiums on that firm's own channel structure, which is the intended use of the procedure in Appendix A. We do not report U for the benchmark collection, because those networks do not satisfy the disjointness assumption and Conjecture 12 is unproved.

For a quarter of the benchmark instances in the public-attribute regime (24 of 96) the measured value is exactly zero, so for those the price this planner could justify is zero. Whether an exact optimiser could justify more is not measured: by Proposition 1, $H = 0$ says the planner returned plans of equal value with and without θ, not that the information changes no decision, and the expected-loss enumeration of Section 7.8 finds $H <$ EVPI on 38% of instances.

7.8 How far the measured quantity sits from EVPI

Proposition 1 is an identity and by itself says only that the sign of $H -$ EVPI is undetermined. We therefore measured the two gaps directly on 10,000 miniature instances across five seeds. Every quantity there, namely EVPI, ε_π, ε_θ and H, is computed by exhaustive enumeration, with the heuristic being the same cost-budgeted greedy rule used in the network pipeline. The identity was verified numerically: the largest residual $|H - (\text{EVPI} + \varepsilon_\pi - \varepsilon_\theta)|$ over all instances was 2.8×10^{-17}.

Table 12 reports the result. The sign of the discrepancy is not stable. Although the mean of $H -$ EVPI is positive, H is strictly *below* EVPI more often than above it (0.38 against 0.15, with the two equal on 48% of instances), and the positive mean comes from a longer right tail rather than from a majority. The direction also moves with the number of focal categories: the mean discrepancy falls from $+0.0008$ at $K = 2$ to -0.0001 at $K = 4$, and $P(H <$ EVPI$)$ rises from 0.12 to 0.63 over the same range (Figure 6, right). That is three points, on instances small enough to enumerate, and we do not extrapolate it in either direction. It is enough to prevent us from assuming the sign is favourable at the scale of the benchmark networks, whose focal-category counts run from 2 to 672 (Section 7.1). We had predicted the opposite before running

Table 12 Planner gaps and the discrepancy between H and EVPI; five seeds, 2000 instances each.

quantity	value
mean EVPI	0.0436
mean H	0.0440
mean ε_π / mean ε_θ	0.00241 / 0.00205
$P(\varepsilon_\pi = 0)$ / $P(\varepsilon_\theta = 0)$	0.83 / 0.54
$P(H = \text{EVPI})$ exactly	0.477
$P(H > \text{EVPI})$ / $P(H < \text{EVPI})$	0.148 / 0.376
mean $H - \text{EVPI}$	+0.00036
mean $\lvert H - \text{EVPI}\rvert$ as a fraction of mean EVPI	0.083
5th, 95th percentile of $H - \text{EVPI}$	−0.0091, +0.0148
Spearman (H, EVPI)	0.971

the experiment, on the reasoning that a planner handed θ faces an easier problem and would more often be exactly optimal. That reasoning is wrong: the prior-side greedy is exactly optimal on 83% of instances and the oracle-side greedy on only 54%, so ε_θ exceeds ε_π more often than the reverse (37.6% against 14.8%, with the two equal on the rest).

Second, the discrepancy is small. It is exactly zero on almost half the instances, and its mean absolute size is about 8% of the mean EVPI. Third, H is a faithful ordering: Spearman correlation with EVPI is 0.971. So H misstates the level modestly and in an unstable direction, while ranking instances almost exactly as EVPI does. One qualification limits how far this carries. Following the convention of Section 3.1, the enumeration of this subsection is performed under expected loss, while the measurements of Sections 7.3–7.6 are taken under CVaR. Its three findings are therefore properties of the expected-loss planner pair, and their transfer to the CVaR measurements is an assumption rather than a measurement; closing it would require re-running the enumeration under CVaR, which we have not done. Subject to that, comparisons across instances in Sections 7.3–7.6 rest on the ordering; statements about the level are reported as planner-specific throughout, and Section 7.7 prices only what was measured.

7.9 Pre-specified tests

Predictions were registered before the corresponding data existed, and two outcomes bear on the paper. A pre-specified test of rank-based predictive validity for the screening score did not meet its reliability criterion, the target's split-half rank reliability being at noise level. A pre-specified moderator hypothesis, that value rises with the ground-truth chokepoint census, was not supported; the outcome is uninformative rather than negative, since the target's reliability was too low for any predictor to have passed. The pre-specified criteria are stated in the header of each analysis script, before any output it produces.

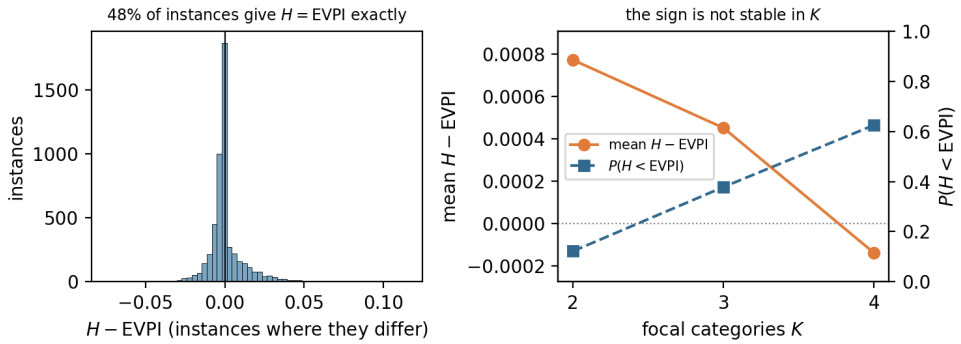


Fig. 6 Left: distribution of $H - \text{EVPI}$ on instances where the two differ. Right: the mean discrepancy and the frequency with which H understates EVPI, against the number of focal categories.

8 Scope and limitations

(1) The measured quantity is H, not EVPI. Section 7.8 measures the gap on miniature instances: the discrepancy is exactly zero on about half of them and averages some 8% of EVPI in absolute size, and H orders instances almost exactly as EVPI does. Its sign is not stable, however, and over the enumerated range $K = 2$ to 4 it trends towards understatement as the number of focal categories grows. The benchmark networks have many more categories than the instances we could enumerate, so we do not claim the reported values bound EVPI from either side. The calibration was also carried out under expected loss while the reported measurements are under CVaR, so its transfer across objectives is assumed rather than shown.

(2) The latent layers are model-generated. The topologies are real; supplier identities, risk attributes, probe outcomes and disruption probabilities are not. The focal layer is additionally single-sourced by construction, so identification can reach the loss only through tier two and deeper (Section 7.1).

(3) Proposition 10 has no empirical illustration: on a fixed instance set the mean advantage does not depend monotonically on λ. The proposition concerns a bound and is not contradicted, but neither is it supported by measurement.

(4) Conjecture 12 is open, and the CVaR version of Theorem 6 does not follow from the expected-loss version.

(5) The bound U is never evaluated on any network in this paper. It is verified by enumeration on miniature instances and it defines the ceiling price of Section 7.7, but the benchmark topologies violate the disjointness assumption of Theorem 6 and Conjecture 12 is unproved, so the one quantity a firm could act on is demonstrated only at miniature scale.

(6) The screening score has a verified bound property factor by factor on disjoint miniature instances and moderate rank agreement with exact EVPPI there. It has no demonstrated validity at network level and none on the benchmark collection.

(7) The mitigation budget is fixed at $B = 9.0$ throughout and never varied. Theorem 6 localises everything to budget breakpoints and Remark 1 shows a category's value depends on whether the consumer cut is affordable, so the measured advantages

could in principle be an artefact of one budget sitting above most cut costs. A sweep over B is the sensitivity our own theory motivates most strongly, with a prediction attached: value should be largest where B falls between securing costs and cut costs. We did not run it.

(8) The determinant of the instances with positive value is unidentified. The census of Section 7.3 rules out the checkable witness of Proposition 8 as the explanation and points instead at candidate multiplicity, which is a feature of this benchmark collection. K_0 itself was tested post-hoc (Section 7.6). At chain level it is associated with the advantage at $\lambda = 0$ about as strongly as the chokepoint census is, and the two are close to collinear here, so the paper reports one marginal association rather than a determinant. The apparent rise of the K_0-scaled advantage with K_0 at $\lambda = 0.8$ does not survive clustering and is not claimed.

(9) Three analyses remain undone: the expected-loss rerun (Section 5.3), the enumeration under CVaR (Section 7.8) and the budget sweep of item 7. Each is a real run rather than a re-aggregation. The paired sweep analysis and the K_0 scaling, which were re-aggregations, have been carried out and are reported in Sections 7.5 and 7.6; the within-instance trend regression of Section 7.5 has not.

(10) All results concern one mitigation problem: budgeted backup qualification under a conjunctive cascade. Inventory buffering, dual sourcing and contractual remedies may price structural information differently. Concurrent dual sourcing in particular is outside the model, as Section 3 notes: the candidate set is epistemic, and no category is served by two suppliers at once.

9 Conclusion

When the candidates of every supply category are exchangeable by a symmetry of the public network that carries their consumers, upstream subtrees and correlation clusters, *and* identification-free actions of no greater cost uniformly dominate, the decision value of hidden supply-network structure is exactly zero, and by Lemma 2 no information-acquisition policy can recover it. The statement concerns the decision problem rather than any method, and it holds for any monotone law-invariant objective, including CVaR. Where those conditions fail, the value is bounded above, for node-disjoint categories, by identification premiums computable in advance from prior and public data, and the bound is attained in a sub-case of a dilution special case. On benchmark multi-echelon topologies with model-generated latent overlays, the advantage that perfect structural information confers on an implemented planner is small under the loss normalisation used here, with a median of exactly zero in the public-attribute regime; positive-value cases are uncommon in the evaluated benchmark set, although their structural determinants remain unresolved. That advantage is a planner-specific measurement whose relation to EVPI is an identity with a sign we do not establish (Proposition 1), and on the miniature instances where both can be computed the discrepancy trends towards understatement over the enumerated range $K = 2$ to 4. The measurements should not be read as bounding the value of structural information from either side.

The practical reading is that the value can be examined in advance. Before commissioning sub-tier discovery, a firm can ask whether its situation satisfies conditions under which the answer is provably nothing, and if not, what the largest possible payoff would be. Verifying (H1) through (H1′) is a search for automorphisms of the public structure, not a comparison of the candidates' marginal risks, costs or prior weights, and it is global: a single category whose candidates differ in their consumers, upstream subtrees or clusters defeats it. Verifying (H2) requires an argument over the plan space, for which we give no general procedure, and its cost clause is a real restriction rather than a formality: in the parameterisation of Definition 2 the consumer cut costs $c_v > c_u$ and securing every candidate costs mc_u, so both canonical identification-free replacements are dearer than the identification-dependent action they replace and (H2) fails. Within this model, therefore, (H2) is witnessed only where a consumer-level securing action is no dearer than securing a named candidate, the opposite of the ordering $c_v > c_u$ used in Definition 2. The in-model example is a shared downstream stage that can be dual-sourced or buffered once. A contractual remedy priced per category rather than per supplier would also serve, but it is not an element of V and so would require an extension of the action space that we do not model. Proposition 8 gives a third route, checkable from public data, and the enumerated batch of Table 4 instantiates (H1) and (H2) by construction at miniature scale. No benchmark network in Section 7 is shown to satisfy them. Where no such action exists, the theorem is silent and Theorem 6 is the relevant statement.

Declarations

Competing interests.

The author declares no competing interests.

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Data and code availability

The Willems [30] data is a public INFORMS/MSOM supplement (DOI 10.1287/msom.1070.0176) and is not redistributed by us. The experiment pipeline, the verifier, the screening-score code, the archived result files from which every reported measurement was computed, the console output of each reported run and the figure scripts are available at https://github.com/rahul-k-mishra/supply-network-information-value, together with a manifest giving the SHA-256 digest of every file in it. Every reported number regenerates deterministically from the archived result files at the pinned revision of that repository, together with the public dataset. Re-executing the experiments from scratch reproduces the reported aggregate statistics but not necessarily every instance-level value, since a change of code revision can shift the pseudorandom stream.

Appendix A The screening score

Theorem 6 suggests a quantity a firm could evaluate on its own network before commissioning discovery work. For each latent channel (v, k) feeding a set $F(v, k)$ of focal categories, let $w = K_0^{-1} \sum_{f \in F} \min\{1, \bar{p}_f/(1-\alpha)\}$ with $\bar{p}_f$ the prior starvation probability of focal category f, the per-category term supplied by Proposition 3; let ϕ be the increase in $K_0^{-1} \sum_{f \in F} \bar{p}_f$ when the channel is forced inoperative in the prior network and everything else is left as it is, one evaluation of the prior model; and let I be the premium (7) on the channel's finite breakpoint set. The *structural information screening score* is

$$\mathrm{SISS} = \sum_{(v,k)} w \cdot \phi \cdot I. \tag{A1}$$

The score is placed here rather than in the body because nothing in the paper depends on it. Each factor I bounds its own channel's partial value by Corollary 9, but the aggregate applies the weight and attribution outside the assumptions of Theorem 6, in particular to channels feeding several focal categories, and we have no theorem that the aggregate remains an upper bound there. The scales also differ: w is a CVaR-scale importance factor while ϕ and I are expected-loss quantities, so the product is on the scale of neither objective. A pre-specified test of its rank-based predictive validity did not meet its reliability criterion (Section 7.9). We therefore report it as a ranking device with no demonstrated validity at network level, and use it nowhere in Sections 6–7.7.

Appendix B Reproducibility and provenance

Data.

Willems [30], public INFORMS/MSOM supplement, DOI 10.1287/msom.1070.0176. Retained networks and the preprocessing rule are given in Table 6.

Determinism.

Seeds derive from a stable checksum of the instance identifier rather than from any hash whose salt varies between processes, so every reported figure regenerates exactly from a fixed code revision. Across revisions the aggregate statistics are reproduced but instance-level values need not be, since a change of revision can shift the pseudo-random stream. The verification pipeline reproduced identical reported outputs on an independent machine.

Frozen result files.

All reported measurements derive from result files archived at submission, with SHA-256 digests

- benchmark, $\lambda = 0$:
  `2f9e68832da1340dbc3fbd9016f87170e6807faa935e44b0bfcf256c93186486`
- synthetic, $\lambda = 0$:
  `39a95a658dc0c6035c62f51a18c46048cc3fc8e037c014e085c44dfb609dcbfd`
- benchmark, $\lambda = 0.8$:
  `2824015b4b0cf6cd1ada407624e4203c048dbaed2947bb4b2efc017328a30f23`

- synthetic, $\lambda = 0.8$:
  `39282f598119d9a9b7e9466e72128c2b61f0e7a2fdf4988abb280a1ea3c0348e`

Regeneration.

One script regenerates every table and figure in Sections 6 and 7 from those files and prints the values that must be reproduced. The enumeration experiments and the dispersion sweep are separate scripts whose pass criteria are stated in their headers before execution.

Appendix C The sufficient condition for (H1)

Throughout, F_v is the failure indicator of node v, operability is defined recursively by $O_v = 1$ if $v \in S$ and $O_v = (1 - F_v) \prod_{k \in K_v} O_{s_{v,k}(\theta)}$ otherwise, where $s_{v,k}(\theta)$ is $\theta_{v,k}$ for a latent category and the public supplier otherwise, and $Z_k = 1 - O_{s_{0,k}(\theta)}$.

Example 1 (Marginal exchangeability is not sufficient) The focal node takes category 1 from w and category 2 from x, both public, so $K_0 = 2$. Node w consumes a latent category a with $C_{w,a} = \{u_1, u_2\}$ and a public category b supplied by u_1; x, u_1 and u_2 have no inputs. Failures are independent, with $q_{u_1} = q_{u_2} = 0.1$, $q_x = 0.15$ and $q_w = 0$. All securing costs equal 1, the budget is $B = 1$, and the prior is uniform. The candidates have the same failure law, cost and prior weight, and differ only in that u_1 also supplies (w, b).
(X1) Under the empty plan $Z_2 = F_x$. If $\theta = u_1$ then $O_w = 1 - F_{u_1}$, so $\mathbb{P}(Z_1 = 1) = 0.1$ and L takes the values $0, \frac{1}{2}, 1$ with probabilities $0.765, 0.220, 0.015$. If $\theta = u_2$ then $O_w = (1 - F_{u_1})(1 - F_{u_2})$, so $\mathbb{P}(Z_1 = 1) = 0.19$ and the probabilities are $0.6885, 0.2830, 0.0285$. The law of L, and even its mean (0.125 against 0.170), depends on θ, so (H1) fails.
(X2) Proposition 8 applies to (w, a): $w \neq 0$, w is the only consumer of u_1 and of u_2, and $c_w \leq \min_u c_u$. Securing w sets $Z_1 = 0$ in every scenario, so it dominates securing either candidate pointwise at no greater cost, and (H2) holds.
(X3) The feasible plans are $\emptyset, \{w\}, \{x\}, \{u_1\}, \{u_2\}$, with $j(\cdot; u_1) = 0.125, 0.075, 0.050, 0.075, 0.125$ and $j(\cdot; u_2) = 0.170, 0.075, 0.095, 0.125, 0.125$. The state-optimal values are 0.050 and 0.075, with mean 0.0625; the prior-optimal plan is $\{x\}$, with value 0.0725. Hence EVPI $= 0.01 > 0$. Step (T5) of Theorem 4 fails: both state-optimal plans are identification-free, but $j(\{x\}; \theta)$ varies with θ.

Variants in which u_1 alone has an upstream supplier that also feeds x, or in which u_1 alone shares a copula cluster with x, keep the mean of L fixed and change its law, so they defeat Corollary 5 while leaving expected loss unchanged.

Lemma 13 (Sufficiency of (H1′)) *Let σ be a structure automorphism. For every plan S and state θ, $Z(\sigma S; \sigma\theta, \cdot)$ has the same law as $\Pi_\sigma Z(S; \theta, \cdot)$, where Π_σ permutes the focal coordinates by σ_0. Consequently (H1′) implies (H1), and the local form stated in Section 4 implies (H1′).*

Proof (Q1) Fix a failure vector F and define G by $G_{\sigma v} = F_v$. The map σ carries the supplier graph of θ onto that of $\sigma\theta$: for a public category this is (A1), and for a latent one $s_{\sigma v, \sigma_v k}(\sigma\theta) = \sigma(\theta_{v,k}) = \sigma(s_{v,k}(\theta))$ by the definition of $\sigma\theta$. Hence σv has the same depth in the graph of $\sigma\theta$ as v in the graph of θ.

(Q2) We show $O_{\sigma v}(\sigma S, \sigma\theta, G) = O_v(S, \theta, F)$ by induction on that depth. Membership agrees because σ is a bijection; own failure agrees because $G_{\sigma v} = F_v$; and by (Q1) and $K_{\sigma v} = \sigma_v(K_v)$ the inputs of σv are the images $\sigma(s_{v,k}(\theta))$, $k \in K_v$, which lie at smaller depth, so the induction hypothesis equates their operability with that of $s_{v,k}(\theta)$. Substituting into the recursion gives the claim; at depth zero the product is empty.
(Q3) By (A0) and (Q1) the supplier of focal category $\sigma_0 k$ under $\sigma\theta$ is $\sigma(s_{0,k}(\theta))$, so (Q2) gives $Z_{\sigma_0 k}(\sigma S, \sigma\theta, G) = Z_k(S, \theta, F)$ for every F.
(Q4) Under the one-factor Gaussian copula, $F_v = \mathbf{1}\{\sqrt{r_g}\, Y_g + \sqrt{1 - r_g}\, \varepsilon_v \le \Phi^{-1}(q_v)\}$ with $g = \kappa(v)$, r_g the cluster's copula parameter, and (Y_g), (ε_v) independent standard normal. By (A3), $G_v = F_{\sigma^{-1}v}$ is the same function of $Y'_g = Y_{\eta^{-1}g}$ and $\varepsilon'_v = \varepsilon_{\sigma^{-1}v}$ that F_v is of Y and ε. These are re-indexings of independent standard normal families by bijections, so G has the law of F. With (Q3) this gives the first statement.
(Q5) Given an identification-free S and θ, θ', take σ from (H1′). Since $\sigma S = S$ and $\sigma\theta = \theta'$, (Q4) gives $Z(S; \theta', \cdot) \overset{d}{=} \Pi_\sigma Z(S; \theta, \cdot)$. The loss (1) is symmetric in the focal coordinates, so $L(S; \theta', \cdot) \overset{d}{=} L(S; \theta, \cdot)$, which is (H1).
(Q6) For the local form, list the latent categories $\ell_1, \ldots, \ell_n$ and set $\theta^{(0)} = \theta$. At step i apply the automorphism τ_i that maps $\theta^{(i-1)}_{\ell_i}$ to θ'_{ℓ_i} (the identity if they agree). It fixes ℓ_i, so the ℓ_i coordinate becomes θ'_{ℓ_i}; it fixes every other latent category together with its candidates, so the other coordinates are unchanged. After n steps the state is θ'. The composition $\tau_n \circ \cdots \circ \tau_1$ satisfies (A0)–(A3), maps θ to θ' and fixes S, which is (H1′). □

The requirement $\sigma S = S$ in (H1′) may be weakened to $L(\sigma S; \theta', \cdot) \overset{d}{=} L(S; \theta', \cdot)$ with the same proof, which matters when an identification-free plan meets two candidates' upstream cones asymmetrically without changing the loss. (H1′) is sufficient, not necessary: (H1) can hold by numerical coincidence without any symmetry. Steps (T5) of Theorem 4 and (C2) of Corollary 5 use only (H1), so both results hold unchanged with (H1′) in place of (H1).

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